

Fares Bentaleb

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EXPERIENCE

Software Engineer Intern | Python, TypeScript

August 2025 – Present

Relevant

Paris, France

- Developing a high-throughput, low-latency virtual try-on API & web app from scratch to generate hyper-personalized images for real-time email marketing campaigns.
- Achieved **<20ms** inference time per image on a single T4 GPU (**>50 FPS**), optimizing AI models (VTOns) for batched processing.
- Engineered an AI-driven feature that translates natural language questions into structured query parameters, populating the UI to construct complex data explorations without manual filter selection.

PROJECTS

Solana MEV Arbitrage Engine

Rust, Solana BPF, gRPC

- Built a real-time circular arbitrage bot that streams live pool state via gRPC, detects N-hop profitable routes across **14 DEX protocols** (AMM, CLMM, DLMM, CLOB), and submits transactions in **sub-millisecond** latency (detect-to-send) on a colocated 5.7GHz server.
- Implemented 256-bit fixed-point swap math (128x128 wide multiply, Knuth's Algorithm D division) for concentrated liquidity pools, achieving **<0.01%** error vs on-chain validator output across **100K+** simulated transactions.
- Designed a parallel nonce fan-out submission system racing **7 providers** simultaneously (Jito bundles, Falcon QUIC, Helius), where the first landed transaction atomically invalidates all others via durable nonce advancement.
- Calibrated **90+ pair-specific** compute unit (on-chain gas) budgets over 16 paper-trading rounds, reaching **85–97%** CU efficiency with zero budget exceeded errors in the final round.

Lumen Protocol | JavaScript, Python

github.com/far3x/lumen-app | lumen.onl

*Architected and deployed a B2B SaaS platform for AI training data, processing **>80M** lines of code from hundreds of users and distributing over **\$40,000** in rewards.*

- Engineered a hybrid valuation engine using pgvector for uniqueness analysis, quantitative metrics, and a multi-stage Gemini API chain for qualitative scoring and open-source verification, completing the full process in under 10 seconds.
- Designed a scalable backend with FastAPI and Celery, utilizing prioritized Redis queues to ensure immediate processing of time-sensitive events via WebSockets.
- Developed a complete B2B SaaS portal and an automated Solana-based payout system to handle on-chain USDC transfers, from secure treasury management to transaction broadcasting.

Pylumen | Python

github.com/far3x/lumen

- Launched an open-source tool with **>7,000** downloads on PyPI for secure data contribution to Lumen.
- Engineered a local-first security pipeline integrating TruffleHog and Scrubadub to programmatically redact secrets and Personally Identifiable Information from codebases before data transmission.
- Built a highly-optimized file traversal system for **>20** languages, handling '.gitignore' rules and smart encoding to generate token-efficient prompts for local AI analysis.

LEADERSHIP

Vice President, Garage ISEP

September 2024 – Present

Student-Run Coding Lab

Paris, France

- Mentor 20+ students weekly in topics from Bash/Git to web development, and led multiple workshops (Git and Docker) for over 50+ students.

EDUCATION

Paris Institute of Digital Technology (ISEP)

Expected July 2027

Master's Degree in Computer Science